6 An electron is confined within a one-dimensional box of width L.

ψ represents the sinusoidal wavefunction of the electron. The graph of $|\psi|^2$ against distance x from one wall of the box is shown in Figure 6.1.

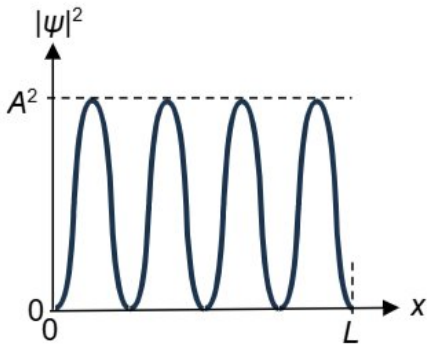


Figure 6.1

(a) (i) State what is represented by $|\psi|^2$.

.....

..... [1]

(ii) The graph of $|\psi|^2$ against x in Figure 6.1 has an amplitude of A^2.

Derive an expression for A in terms of L.

A = ... [2]

(b) The electron with the $|\psi|^2$ graph shown in Figure 6.1 is in an excited quantum state n in a box where L = 0.10 nm.

(i) State the quantum state n of the trapped electron shown in Figure 6.1.

n = ... [1]

(ii) Calculate the momentum of the electron at this quantum state.

momentum =..... N s [2]

(iii) Calculate the energy, in eV, of the electron at this quantum state.

energy = ... eV [2]

[Total: 8]